



M1 Internship

Out-of-distribution detection by model distillation

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1 Introduction

Modern image classifiers assume that test inputs resemble the data used for training, these inputs are called In-Distribution (ID). That assumption fails in open environments. A medical classifier may receive an image from another acquisition protocol, and a driving system may encounter an object or visual condition absent from its training set. These abnormal inputs are called Out-Of-Distribution (OOD). A network can then return a confident prediction even though none of its classes provides a valid description of the input. Out-of-distribution detection adds a scalar score and a rejection rule to identify such cases before the prediction is used.

This internship started from a proposal to use model distillation as an OOD detector, since this area is yet poorly explored in OOD detection related scientific papers. A high-capacity teacher is fixed after training on in-distribution data. A smaller student learns only from those data and tries to reproduce a teacher output or representation. If the student captures regularities that hold on the training distribution, teacher-student disagreement may increase on OOD inputs.

The project also explored two extensions beyond matching the teacher’s class outputs. In the subspace-student ensemble, each student is trained on a fixed or randomly sampled subset of an internal teacher representation. Disagreement across the ensemble measures how predictions change when each student receives only part of the representation. Feature denoising follows a different pipeline and diverges from classical knowledge distillation: a student receives a corrupted teacher feature and learns to reconstruct its clean version. This direction was partly inspired by LeCun’s proposal for predictive representation learning and by joint-embedding predictive architectures (JEPA), which predict representations from incomplete context rather than reconstructing raw pixels [8, 9].

1.1 Progression of OOD detection methods

The exploration of methods to detect OOD inputs has been in progress since 2017. Early work used the final softmax output. Later methods measured distance in representation space, changed the training distribution, or modified internal activations at inference time.

Year	Method	Main idea
2017	Maximum softmax probability (MSP) [2]	Use the largest predicted class probability as an ID-confidence score. The method needs no retraining, but a classifier can remain confident on OOD data.
2018	Mahalanobis distance [3]	Fit class-conditional Gaussian statistics to features from several network layers. Score a test point by its distance to the closest class mean.
2021	ReAct [4]	Clip unusually large penultimate activations before computing a confidence score. The clipping threshold comes from ID data and reduces OOD overconfidence.
2022	Deep k -NN [5]	Normalize deep features and measure the distance to nearby ID training features. This removes the Gaussian assumption of Mahalanobis scoring.
2022	PixMix [6]	Train with random compositions of images and structurally complex mixing images. The method changes training data to improve several safety measures, including anomaly detection.

Table 1: A short timeline of the methods that motivated the internship.

These methods served as inspiration for some of the experimental axes explored here. MSP is a teacher-only baseline. Mahalanobis and k -NN motivate representation-space scores. ReAct motivates activation clipping. PixMix motivates pixel perturbation and feature prediction under image corruption.

1.2 Research questions

The work addressed the following questions.

1. Does a small student disagree with its teacher more often on OOD inputs than on ID inputs?

2. Does corruption during student training expose a stronger signal than clean distillation?
3. Should students reproduce class outputs, reconstruct internal features, or form an ensemble trained on different feature subspaces?
4. How stable are the conclusions across teacher architectures, ID datasets, OOD datasets, score definitions, and evaluation conventions?

2 Distillation framework

2.1 Teacher and student

Classical knowledge distillation compresses a teacher into a smaller student by matching softened class distributions [1]. Let x be an input and let C be the number of classes. The teacher and student produce logit vectors $z_T(x), z_S(x) \in \mathbb{R}^C$, where the subscripts T and S identify the teacher and student. The component $z_{T,c}(x)$ is the teacher logit for class c , and $z_{S,c}(x)$ is defined analogously. For a temperature $\tau > 0$, their softened class probabilities are

$$p_T^{(\tau)}(c | x) = \frac{\exp(z_{T,c}(x)/\tau)}{\sum_{j=1}^C \exp(z_{T,j}(x)/\tau)}, \quad p_S^{(\tau)}(c | x) = \frac{\exp(z_{S,c}(x)/\tau)}{\sum_{j=1}^C \exp(z_{S,j}(x)/\tau)}.$$

Here c is the class whose probability is evaluated and j indexes all C classes in the denominator. A temperature above one exposes relative probabilities among classes that may be almost invisible at $\tau = 1$. Hinton et al. multiply the soft-target loss by τ^2 to compensate for the temperature-dependent gradient scale.

During training, both teacher and student see only ID data. During evaluation, a scalar score, called OOD score, is computed from the teacher, the student, or their difference.

2.2 Training objectives

The project implemented centered-logit mean squared error, soft and hard cross-entropy, and Kullback–Leibler (KL) divergence. To simplify notation, the dependence on x is omitted below. For $M \in \{T, S\}$, define the mean logit $\bar{z}_M = C^{-1} \sum_{c=1}^C z_{M,c}$, and let $\mathbf{1} \in \mathbb{R}^C$ be the vector of ones. The centered-logit loss is

$$\mathcal{L}_{\text{MSE}} = \|(z_S - \bar{z}_S \mathbf{1}) - (z_T - \bar{z}_T \mathbf{1})\|_2^2.$$

The squared Euclidean norm $\|\cdot\|_2^2$ measures the distance between the two centered logit vectors. Centering removes any constant logit offset, which softmax ignores. The mixed cross-entropy objective is

$$\mathcal{L}_{\text{CE}} = \alpha \tau^2 H(p_T^{(\tau)}, p_S^{(\tau)}) + (1 - \alpha) H(y, p_S^{(1)}),$$

where $H(q, p) = -\sum_{c=1}^C q(c) \log p(c)$ is cross-entropy, $\alpha \in [0, 1]$ controls the balance between the soft teacher target and the hard label, and y is the one-hot ID class label. The first term transfers the teacher’s full class distribution; the second retains ordinary supervised classification. We also used the forward KL objective

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(p_T \| p_S).$$

Here $p_T = p_T^{(1)}$ and $p_S = p_S^{(1)}$ unless another shared temperature is specified. The divergence is

$$D_{\text{KL}}(p_T \| p_S) = \sum_{c=1}^C p_T(c) \log \left(\frac{p_T(c)}{p_S(c)} \right).$$

Minimizing it makes the student distribution approach the fixed teacher distribution.

2.3 OOD Scores

OOD Score is oriented so that a larger value means more ID-like. Raw distances, errors, and uncertainties are negated. We evaluated four score families. Teacher-only scores serve as reference baselines.

Family	Examples	Interpretation
Teacher-only baseline	MSP; energy [7]	Reference confidence of the fixed teacher without a student.
Student-only	Student MSP; student energy	Confidence learned by the limited student.
Teacher-student mismatch	$-D_{\text{KL}}(p_T \parallel p_S)$; negative centered-logit distance; signed or absolute MSP and energy gaps	Agreement between teacher and student outputs.
Reconstruction and ensemble	Negative reconstruction error; absolute or relative reconstruction improvement; cosine similarity; negative predictive entropy; negative BALD	Predictability of a clean feature or agreement among students trained on fixed or randomly sampled feature subspaces.

Table 2: OOD Score families implemented during the internship.

3 Methods and experimental protocol

3.1 Models and datasets

The experiments used ResNet-18, ResNet-50, and ViT-B/16 teachers. ResNet students consumed flattened feature maps, globally averaged features, or selected feature coordinates. ViT feature-denoising students used CLS embeddings or patch-token grids. The student family included linear models, multilayer perceptrons, random forests, and residual convolutional denoisers.

The first evaluation used CIFAR-10 and CIFAR-100. The opposite CIFAR dataset was Near-OOD, while MNIST and SVHN were Far-OOD. The last phase adopted the OpenOOD v1.5 benchmark [13]. OpenOOD is an effort to standardize OOD evaluation through fixed datasets, image-list splits, and aggregation groups. Many papers about OOD detection often use different OOD datasets or different splits under the same benchmark name, making their reported metrics impossible to compare directly. OpenOOD v1.5 added Tiny ImageNet (TIN), Textures, and Places365 to the CIFAR protocol, and introduced ImageNet-200 with SSB-hard and NINCO as Near-OOD and iNaturalist, Textures, and OpenImage-O as Far-OOD.

Phase	ID dataset	Near-OOD	Far-OOD
Initial CIFAR	CIFAR-10	CIFAR-100	MNIST, SVHN
Initial CIFAR	CIFAR-100	CIFAR-10	MNIST, SVHN
OpenOOD CIFAR-10	CIFAR-10	CIFAR-100, TIN	MNIST, SVHN, Textures, Places365
OpenOOD CIFAR-100	CIFAR-100	CIFAR-10, TIN	MNIST, SVHN, Textures, Places365
OpenOOD ImageNet	ImageNet-200	SSB-hard, NINCO	iNaturalist, Textures, OpenImage-O

Table 3: Dataset protocols used during the internship.

3.2 Baseline distillation

The baseline student receives a clean teacher feature and predicts teacher logits or probabilities. We compared linear, MLP, and random-forest students. This establishes whether limited capacity alone creates an OOD signal

before any explicit corruption is added.

3.3 Embedding perturbations

Embedding perturbations modify an intermediate teacher activation before the student sees it. The tested operations were percentile clipping, Monte Carlo dropout, PCA projection, and masked PCA projection. Clipping was applied to a single internal teacher layer or sequentially to several internal teacher layers. Its threshold could be global, channel-dependent, or spatially dependent. Training targets were either the clean teacher output or the teacher output after the same perturbation. The student received perturbed feature vectors concatenated with perturbation parameters.

3.4 Pixel perturbations

Pixel perturbation was tested with affine and photometric augmentation and with PixMix. The affine pipeline sampled rotation, translation, scaling, brightness, and contrast. The student received a feature extracted from the perturbed image concatenated with normalized transformation parameters. PixMix repeatedly combined a crop-and-flip view with an augmentation or an external structurally complex image. These experiments tested whether prediction stability under input changes separates ID and OOD examples.

3.5 Subspace-student ensemble

The ensemble trained 16 linear students on channel or whitened PCA subspaces. One variant used fixed ordered partitions; another used seeded random partitions. In both cases, the 512 feature coordinates were divided among the students so that each coordinate was assigned to exactly one student. Predictive entropy and BALD measured uncertainty and disagreement across the 16 predictions [11].

4 Experimental progression

The initial CIFAR experiments explored many combinations of teacher architecture, student model, feature layer, perturbation, training objective, inference protocol, and OOD Score. For brevity, this report retains only the best-performing variant within each method family and ID dataset. Feature-denoising experiments are reported separately in Section 5.

AUROC is the main ranking metric used in OOD detection. Under the ID-positive convention used here, it measures how often a randomly selected ID sample receives a higher score than a randomly selected OOD sample. It is threshold-independent. FPR@95 instead describes a particular operating point: the threshold is chosen to correctly accept 95% of ID samples, and the metric reports the proportion of OOD samples that incorrectly pass that threshold and are accepted as ID. Thus, higher AUROC and lower FPR@95 are better.

4.1 CIFAR-10 as ID

Table 4 reports the selected configuration for each method family.

Method family	Best-performing variant	Selected OOD Score	Near-OOD	Far-OOD			Overall
			CIFAR-100	MNIST	SVHN	Average	Macro
Teacher baseline	ResNet-18; no student	Teacher MSP	0.879 / 0.624	0.918 / 0.556	0.904 / 0.598	0.911 / 0.577	0.900 / 0.593
Teacher baseline	ResNet-50; no student	Teacher MSP	0.873 / 0.628	0.897 / 0.622	0.891 / 0.680	0.894 / 0.651	0.887 / 0.643
Clean distillation	ResNet-18 layer 4; 100-tree random forest; output regression	Student energy	0.870 / 0.550	0.920 / 0.490	0.900 / 0.550	0.910 / 0.520	0.895 / 0.531
MC dropout	ResNet-18 layer 4; channel dropout $p = 0.8$; logit MSE; dropout inference	Student energy	0.885 / 0.498	0.958 / 0.252	0.930 / 0.380	0.944 / 0.316	0.924 / 0.376
Multi-layer clipping	ResNet-18 layers 3–4; channel/flatten; clean target; KL; clipped inference	Student energy	0.859 / 0.581	0.998 / 0.001	0.897 / 0.482	0.948 / 0.242	0.918 / 0.355
Affine pixel perturbation	ResNet-50 layer 4; clean target; logit MSE; clean inference	Student energy	0.871 / 0.552	0.906 / 0.528	0.960 / 0.247	0.933 / 0.388	0.912 / 0.442
Subspace-student ensemble	16 ResNet-18 linear students; ordered disjoint channel-GAP subspaces; KL	Predictive entropy	0.884 / 0.574	0.930 / 0.468	0.905 / 0.554	0.918 / 0.511	0.906 / 0.532

Table 4: Best-performing variant in each method family with CIFAR-10 as ID. Cells report AUROC / FPR@95 on the initial CIFAR protocol.

4.2 CIFAR-100 as ID

Table 5 repeats the selection independently with CIFAR-100 as ID.

Method family	Best-performing variant	Selected OOD Score	Near-OOD	Far-OOD			Overall
			CIFAR-10	MNIST	SVHN	Average	Macro
Teacher baseline	ResNet-18; no student	Teacher MSP	0.792 / 0.794	0.701 / 0.956	0.806 / 0.812	0.754 / 0.884	0.767 / 0.854
Teacher baseline	ResNet-50; no student	Teacher MSP	0.792 / 0.781	0.811 / 0.823	0.758 / 0.793	0.785 / 0.808	0.787 / 0.799
Clean distillation	ResNet-18 layer 4; 100-tree random forest; output regression	Student energy	0.790 / 0.810	0.710 / 0.940	0.840 / 0.630	0.775 / 0.785	0.780 / 0.793
MC dropout	ResNet-18 layer 4; spatial dropout $p = 0.8$; KL; clean inference	KL mismatch	0.768 / 0.881	0.772 / 0.821	0.812 / 0.854	0.792 / 0.838	0.784 / 0.852
Multi-layer clipping	ResNet-18 layers 3–4; channel/flatten; clean target; KL; clean inference	Student energy	0.751 / 0.842	0.797 / 0.879	0.940 / 0.340	0.869 / 0.610	0.829 / 0.687
Affine pixel perturbation	ResNet-50 layer 4; clean target; logit MSE; clean inference	Student energy	0.779 / 0.797	0.875 / 0.698	0.925 / 0.431	0.900 / 0.565	0.860 / 0.642
Subspace-student ensemble	16 ResNet-18 linear students; ordered disjoint flattened-channel subspaces; logit MSE	Predictive entropy	0.779 / 0.818	0.760 / 0.947	0.874 / 0.688	0.817 / 0.818	0.804 / 0.818

Table 5: Best-performing variant in each method family with CIFAR-100 as ID. Cells report AUROC / FPR@95 on the initial CIFAR protocol.

Across both ID datasets, no student family dominated every setting. The selected variant changed with the ID dataset, and strong AUROC did not always coincide with low FPR@95. This instability motivated the later emphasis on feature reconstruction pipeline.

5 Feature denoising

5.1 Motivation

Feature denoising is not classical logit distillation. It follows the predictive-representation idea associated with joint-embedding predictive architectures: predict a target representation from a partial or corrupted context instead of reconstructing raw pixels [8, 9]. Here the teacher remains fixed. A small student learns an ID-conditional prediction rule in the teacher’s feature space. An OOD input should be harder to reconstruct because its feature relations were absent from student training.

For a clean teacher feature z , a corruption produces $\tilde{z}$. The student predicts $\hat{z} = S(\tilde{z})$ and minimizes an MSE on all or only hidden coordinates. We used three reconstruction scores:

$$\begin{aligned} s_{\text{rec}} &= -\|\hat{z} - z\|_2^2, \\ s_{\text{abs}} &= \|\tilde{z} - z\|_2^2 - \|\hat{z} - z\|_2^2, \\ s_{\text{cos}} &= \frac{\hat{z}^\top z}{\|\hat{z}\|_2 \|z\|_2}. \end{aligned}$$

These are respectively the negative reconstruction error, the absolute improvement over the zero-filled or otherwise corrupted input, and cosine similarity. The results below use absolute improvement because it directly measures how much the trained denoiser corrects the corruption.

5.2 Tested variants

The experiments predicted clean pooled embeddings from affine-augmented, PixMix, or block-masked images, and reconstructed masked channels, masked spatial positions, or ViT patch tokens. ResNet feature-map denoisers used residual CNNs. The ViT patch-token model received a 14×14 grid of 768-dimensional layer-12 tokens and used three 3×3 convolutions $768 \rightarrow 64 \rightarrow 64 \rightarrow 768$. A Bernoulli mask hid 20% of patch tokens, and the loss was evaluated only at hidden positions.

Variant	Corruption	Student target	Score candidates
Pixel-to-embedding	Affine or PixMix image	Clean pooled feature	Negative reconstruction error, absolute improvement, cosine similarity
Channel masking	Hidden feature-map channels	Clean feature map	Negative reconstruction error, absolute improvement
Spatial masking	Hidden positions or blocks	Clean feature map	Negative reconstruction error, absolute improvement
ViT patch masking	Hidden layer-12 patch tokens	Clean patch-token grid	Negative reconstruction error, absolute improvement

Table 6: Feature-denoising variants retained in this report.

5.3 Layerwise results on the initial CIFAR protocol

For the ResNet-18 channel-masking pipeline, each channel was independently hidden with probability 0.2, and a residual CNN reconstructed the clean feature map. We evaluated the performance on several internal teacher layers: layers 2, 3, and 4. Layer 3 was strongest overall for CIFAR-10 as ID and gave particularly strong SVHN separation; layer 4 was strongest on MNIST. With CIFAR-100 as ID, layer 3 gave the best compromise, despite remaining weak on Near-OOD.

ID dataset	Layer	MNIST	SVHN	Near-OOD CIFAR	Macro
CIFAR-10	2	0.597 / 0.918	0.937 / 0.271	0.542 / 0.913	0.655 / 0.754
CIFAR-10	3	0.846 / 0.835	0.962 / 0.217	0.841 / 0.662	0.872 / 0.594
CIFAR-10	4	0.979 / 0.115	0.893 / 0.458	0.773 / 0.740	0.855 / 0.513
CIFAR-100	2	0.881 / 0.545	0.900 / 0.421	0.552 / 0.950	0.721 / 0.716
CIFAR-100	3	0.994 / 0.022	0.936 / 0.283	0.504 / 0.967	0.735 / 0.560
CIFAR-100	4	0.621 / 0.981	0.806 / 0.760	0.774 / 0.825	0.744 / 0.848

Table 7: Residual-CNN results with uniform 20% channel masking and absolute improvement as the OOD Score. Cells report AUROC / FPR@95 under the original ID-positive convention.

5.4 Combining layer scores

Since layer 3 performed better on Far-OOD and layer 4 performed better on Near-OOD, it motivated a composition of their absolute-improvement scores. This experiment was inspired by HEAT, which composes several energy terms through a generalized energy operator [10]. We standardized each layer score using its mean and population standard deviation on ID data, then computed

$$s_{\beta}(z_3, z_4) = \frac{1}{\beta} \log(e^{\beta z_3} + e^{\beta z_4}), \quad \beta \neq 0, \quad s_0 = \frac{z_3 + z_4}{2}.$$

Negative β emphasizes the smaller standardized score, positive β emphasizes the larger score, and values near zero produce a smoother balance.

β	CIFAR-10 ID macro	CIFAR-100 ID macro
-100	0.893 / 0.513	0.830 / 0.547
-10	0.895 / 0.511	0.830 / 0.546
-1	0.916 / 0.400	0.836 / 0.512
-0.1	0.920 / 0.359	0.831 / 0.502
0	0.920 / 0.359	0.829 / 0.505
0.1	0.920 / 0.358	0.827 / 0.510
1	0.917 / 0.356	0.802 / 0.581
10	0.902 / 0.424	0.762 / 0.697
100	0.901 / 0.440	0.760 / 0.704

Table 8: HEAT-inspired composition of ResNet-18 layer-3 and layer-4 absolute-improvement scores. Cells report balanced-macro AUROC / FPR@95 under the ID-positive convention.

The useful range was concentrated near zero. With CIFAR-10 as ID, $\beta = 0.1$ reached 0.920 / 0.358; with CIFAR-100 as ID, $\beta = -0.1$ reached 0.831 / 0.502, a better performance than on each layer separately without the composition.

5.5 ViT patch-token denoising

We tested ViT patch-token denoising quickly during the final days of the internship, after moving to the OpenOOD-inspired evaluation described in Section 6. At that point, we also changed the metric convention from ID-positive to OOD-positive so that the evaluation matched the OpenOOD benchmark convention.

Patch-token denoising remained below the teacher-only ViT MSP baseline. The table reports absolute improvement for the denoiser, consistently with the ResNet reconstruction results above.

Table 9: CIFAR-10 as the ID dataset. ViT patch-token denoising and the teacher-only MSP baseline on OpenOOD v1.5; cells report AUROC / FPR@95 under the stated positive-class convention.

Variant	Conv.	OOD Score	Near-OOD			Far-OOD					Balanced
			CIFAR-100	TIN	Average	MNIST	Places365	SVHN	Textures	Average	Macro
Teacher ViT-B/16	ID+	MSP	0.971 / 0.124	–	–	0.993 / 0.024	–	0.991 / 0.035	–	–	0.985 / 0.061
Teacher ViT-B/16	OOD+	MSP	0.971 / 0.138	0.991 / 0.042	0.981 / 0.090	0.995 / 0.020	0.995 / 0.024	0.991 / 0.041	0.999 / 0.0003	0.995 / 0.021	0.988 / 0.056
Patch-token denoising L12	OOD+	Absolute improvement	0.863 / 0.673	0.942 / 0.386	0.903 / 0.529	0.429 / 0.920	0.964 / 0.215	0.919 / 0.430	0.996 / 0.012	0.827 / 0.394	0.865 / 0.462

Table 10: CIFAR-100 as the ID dataset. ViT patch-token denoising and the teacher-only MSP baseline on OpenOOD v1.5; cells report AUROC / FPR@95 under the stated positive-class convention.

Variant	Conv.	OOD Score	Near-OOD			Far-OOD					Balanced
			CIFAR-100	TIN	Average	MNIST	Places365	SVHN	Textures	Average	Macro
Teacher ViT-B/16	ID+	MSP	0.926 / 0.375	–	–	0.684 / 0.845	–	0.930 / 0.414	–	–	0.847 / 0.545
Teacher ViT-B/16	OOD+	MSP	0.926 / 0.316	0.938 / 0.238	0.932 / 0.277	0.689 / 0.662	0.946 / 0.217	0.932 / 0.255	0.988 / 0.076	0.889 / 0.303	0.911 / 0.290
Patch-token denoising L12	OOD+	Absolute improvement	0.629 / 0.889	0.772 / 0.806	0.701 / 0.847	0.148 / 0.979	0.822 / 0.717	0.644 / 0.812	0.935 / 0.340	0.638 / 0.712	0.669 / 0.780

6 Transition to fixed OpenOOD evaluation

Near the end of the internship, we replaced the evaluation based on standard `torchvision` test splits with the fixed OpenOOD v1.5 image-list manifests [12, 13]. As summarized in Table 3, this transition added the benchmark’s Near-OOD and Far-OOD datasets and fixed the exact images used for every comparison. For CIFAR-10 and CIFAR-100, the manifests add TIN as Near-OOD and Textures and Places365 as Far-OOD. For ImageNet-200, they specify SSB-hard and NINCO as Near-OOD and curated subsets of iNaturalist, Textures, and OpenImage-O as Far-OOD. This standardized protocol prevents accidental changes in dataset size and makes results more directly comparable with published OpenOOD values.

While moving to this evaluation, we found that the metric convention was still not fully standardized. In particular, papers and even different OpenOOD artifacts alternate between treating ID or OOD samples as the positive class. The convention does not change AUROC when labels and score direction are reversed together, but it changes the operating point used by FPR@95. One convention can therefore produce a substantially lower FPR@95 for exactly the same predictions without representing a better detector.

6.1 Two definitions of FPR@95

Let s_{ID} increase with ID-likeness and let $s_{\text{OOD}} = -s_{\text{ID}}$. Two different operating points are called FPR@95 in the literature.

Name	Positive / negative	Threshold condition	Reported false-positive rate
ID+	ID / OOD	$\Pr(s_{\text{ID}} \geq \tau_{\text{ID}} \mid \text{ID}) = 0.95$	$\text{FPR}_{95\text{ID}+} = \Pr(s_{\text{ID}} \geq \tau_{\text{ID}} \mid \text{OOD})$
OOD+	OOD / ID	$\Pr(s_{\text{OOD}} \geq \tau_{\text{OOD}} \mid \text{OOD}) = 0.95$	$\text{FPR}_{95\text{OOD}+} = \Pr(s_{\text{OOD}} \geq \tau_{\text{OOD}} \mid \text{ID})$

Table 11: The two operating points commonly called FPR@95.

Under ID+, FPR@95 is the share of OOD inputs accepted as ID while 95% of ID inputs are retained. Under OOD+, it is the share of ID inputs flagged as OOD while 95% of OOD inputs are detected. They use different thresholds and are not complements.

AUROC is not influenced by the convention because it measures ranking. Let X_{ID} and X_{OOD} be independently sampled ID and OOD examples. Under ID+, a score s_{ID} increases with ID-likeness,

$$\text{AUROC}_{\text{ID}+} = \Pr(s_{\text{ID}}(X_{\text{ID}}) > s_{\text{ID}}(X_{\text{OOD}})).$$

Under OOD+, the score direction is reversed, $s_{\text{OOD}} = -s_{\text{ID}}$, and the OOD example is the positive one. Therefore,

$$\begin{aligned} \text{AUROC}_{\text{OOD}+} &= \Pr(s_{\text{OOD}}(X_{\text{OOD}}) > s_{\text{OOD}}(X_{\text{ID}})) \\ &= \Pr(-s_{\text{ID}}(X_{\text{OOD}}) > -s_{\text{ID}}(X_{\text{ID}})) \\ &= \Pr(s_{\text{ID}}(X_{\text{ID}}) > s_{\text{ID}}(X_{\text{OOD}})) = \text{AUROC}_{\text{ID}+}. \end{aligned}$$

6.2 OpenOOD version mismatch

OpenOOD cannot be assigned one convention without naming the artifact and date. Its v1.0 paper used ID+. The v1.5 paper and published spreadsheet use OOD+, but the tagged v1.5 evaluator still encoded ID as positive. The evaluator on the later `main` branch switched to OOD as positive. This history explains why our early project values and the OpenOOD spreadsheet could have matching AUROC but different FPR@95.

We verified the spreadsheet convention with the three released CIFAR-10 ResNet-18 seeds. Mean AUROC was 87.16 under either simultaneous label and score reversal. Mean $\text{FPR}_{95\text{ID}+}$ was 61.01, while mean $\text{FPR}_{95\text{OOD}+}$ was 53.12. The spreadsheet reported 53.08 ± 4.86 , which matches the OOD-positive calculation. Thus, the apparently lower FPR@95 came from evaluating a different operating point, not from improved predictions.

Source	Convention	Consequence
OpenOOD v1.0 paper	ID+	–
OpenOOD v1.5 paper	OOD+	–
GitHub tag v1.5	ID+	Tagged code disagrees with the v1.5 paper.
Later and current main	OOD+	Code follows the v1.5 paper.
Published v1.5 spreadsheet	OOD+	Reference numbers match the OOD-positive calculation.

Table 12: OpenOOD convention timeline.

6.3 The problem persists in recent work

An audit of representative and recent papers found both conventions. Most of the classical methods used ID+, sometimes only implicitly. Several recent methods and the later OpenOOD evaluator used OOD+. Some papers did not state enough information to decide.

Paper or method	Convention
MSP [2]	ID+
Mahalanobis [3]	ID+, implicit
Deep k -NN [5]	ID+
PixMix [6]	Not specified
ReAct [4]	ID+
ASH [14]	ID+
InterNeg, CVPR 2026 [17]	OOD+
Extended Logit Normalization, CVPR 2026 [18]	OOD+
RankOOD, CVPR 2026 [19]	ID+

Table 13: Examples from the convention audit in different papers. Recent publication date does not resolve the ambiguity.

6.4 Full results under both conventions

Tables 14–16 reproduce the full OpenOOD comparisons from the final experiment recap. Every cell reports AUROC/FPR@95 as a fraction. Blue variant cells denote ID+ and red variant cells denote OOD+. Missing entries correspond to datasets that were not part of the earlier, non-OpenOOD evaluation.

Variant	Conv.	Selected OOD score	Near-OOD			Far-OOD					Balanced
			CIFAR-100	TIN	Average	MNIST	Places365	SVHN	Textures	Average	Macro
Teacher MSP (R18)	ID+	MSP	.879/.624	–	–	.918/.556	–	.904/.598	–	–	.895/.601
Teacher MSP (R18)	OOD+	MSP	.880/.437	.891/.378	.885/.408	.925/.230	.894/.373	.918/.266	.885/.412	.906/.321	.896/.364
Teacher energy (R18)	ID+	Energy	.879/.506	–	–	.958/.260	–	.918/.412	–	–	.908/.421
Teacher energy (R18)	OOD+	Energy	.879/.541	.897/.461	.888/.501	.963/.159	.909/.418	.935/.298	.879/.549	.921/.356	.905/.429
Teacher MSP (R50)	ID+	MSP	.873/.628	–	–	.897/.622	–	.891/.680	–	–	.884/.639
Teacher MSP (R50)	OOD+	MSP	.873/.472	.889/.382	.881/.427	.920/.237	.895/.352	.889/.304	.876/.417	.895/.328	.888/.377
Teacher energy (R50)	ID+	Energy	.868/.517	–	–	.932/.408	–	.911/.517	–	–	.895/.490
Teacher energy (R50)	OOD+	Energy	.868/.598	.896/.475	.882/.536	.955/.176	.913/.392	.908/.325	.868/.539	.911/.358	.897/.447
Affine pixels (R50)	ID+	Student energy	.871/.552	–	–	.906/.528	–	.960/.247	–	–	.902/.469
Affine pixels (R50)	OOD+	Student energy	.871/.565	.891/.468	.881/.517	.931/.255	.905/.400	.961/.146	.847/.619	.911/.355	.896/.436
Channel clipping (R18)	ID+	Student energy	.843/.554	–	–	.946/.288	–	.910/.424	–	–	.886/.455
Channel clipping (R18)	OOD+	Student energy	.844/.639	.878/.544	.861/.592	.937/.302	.880/.547	.907/.480	.893/.508	.904/.459	.883/.525
Feature denoising L2 (R18)	ID+	Absolute improvement	.542/.913	–	–	.597/.918	–	.937/.271	–	–	.655/.754
Feature denoising L2 (R18)	OOD+	Absolute improvement	.541/.953	.599/.896	.570/.924	.753/.652	.590/.901	.958/.211	.647/.993	.737/.689	.653/.807
Feature denoising L3 (R18)	ID+	Absolute improvement	.841/.662	–	–	.846/.835	–	.962/.217	–	–	.872/.594
Feature denoising L3 (R18)	OOD+	Absolute improvement	.840/.548	.895/.374	.868/.461	.880/.309	.905/.330	.975/.123	.930/.346	.922/.277	.895/.369
Feature denoising L4 (R18)	ID+	Absolute improvement	.773/.740	–	–	.979/.115	–	.893/.458	–	–	.854/.514
Feature denoising L4 (R18)	OOD+	Absolute improvement	.772/.721	.778/.704	.775/.713	.980/.100	.803/.648	.912/.413	.751/.793	.862/.489	.818/.601
Mahalanobis [3]	OOD+	MDS	.836/.528	.848/.470	.842/.499	.901/.273	.849/.477	.912/.260	.927/.279	.897/.322	.870/.411
k-NN [5]	OOD+	KNN	.897/.376	.916/.304	.906/.340	.943/.201	.918/.304	.927/.226	.932/.241	.930/.243	.918/.291
PixMix [6]	OOD+	MSP	.919/.289	.943/.210	.931/.250	.917/.274	.941/.211	.979/.101	.989/.066	.957/.163	.944/.206
ReAct [4]	OOD+	ReAct	.859/.674	.883/.597	.871/.636	.928/.338	.903/.442	.891/.502	.894/.514	.904/.449	.888/.542
ASH [14]	OOD+	ASH	.741/.873	.764/.863	.753/.868	.832/.700	.799/.779	.735/.836	.775/.846	.785/.790	.769/.829
ViM [15]	OOD+	ViM	.877/.492	.896/.405	.887/.448	.948/.184	.895/.414	.945/.193	.952/.211	.935/.251	.911/.349

Table 14: CIFAR-10 as the ID dataset. Cells report AUROC/FPR@95 under the stated positive-class convention.

Variant	Conv. Selected OOD score		Near-OOD			Far-OOD					Balanced
			CIFAR-10	TIN	Average	MNIST	Places365	SVHN	Textures	Average	Macro
Teacher MSP (R18)	ID+	MSP	.792/.794	–	–	.701/.956	–	.806/.812	–	–	.773/.839
Teacher MSP (R18)	OOD+	MSP	.790/.593	.833/.484	.811/.538	.776/.556	.802/.545	.795/.545	.781/.603	.788/.562	.800/.550
Teacher energy (R18)	ID+	Energy	.795/.800	–	–	.695/.974	–	.829/.771	–	–	.778/.836
Teacher energy (R18)	OOD+	Energy	.793/.597	.835/.483	.814/.540	.780/.553	.801/.554	.816/.521	.789/.605	.796/.558	.805/.549
Teacher MSP (R50)	ID+	MSP	.792/.781	–	–	.811/.823	–	.758/.793	–	–	.788/.795
Teacher MSP (R50)	OOD+	MSP	.790/.618	.819/.541	.804/.580	.820/.437	.786/.632	.741/.729	.766/.675	.778/.618	.791/.599
Teacher energy (R50)	ID+	Energy	.801/.774	–	–	.878/.670	–	.783/.766	–	–	.816/.746
Teacher energy (R50)	OOD+	Energy	.799/.645	.828/.580	.813/.613	.878/.366	.781/.705	.773/.712	.779/.686	.803/.617	.808/.615
Affine pixels (R50)	ID+	Student energy	.779/.797	–	–	.875/.698	–	.925/.431	–	–	.839/.681
Affine pixels (R50)	OOD+	Student energy	.777/.715	.817/.596	.797/.656	.886/.334	.770/.716	.929/.248	.797/.623	.846/.480	.821/.568
Channel clipping (R18)	ID+	Student energy	.751/.842	–	–	.797/.879	–	.940/.340	–	–	.810/.726
Channel clipping (R18)	OOD+	Student energy	.749/.699	.829/.508	.789/.604	.865/.392	.807/.575	.937/.256	.858/.547	.867/.442	.828/.523
Feature denoising L2 (R18)	ID+	Absolute improvement	.552/.950	–	–	.881/.545	–	.900/.421	–	–	.721/.717
Feature denoising L2 (R18)	OOD+	Absolute improvement	.553/.894	.536/.898	.545/.896	.916/.322	.571/.879	.914/.372	.408/.999	.702/.643	.623/.770
Feature denoising L3 (R18)	ID+	Absolute improvement	.504/.967	–	–	.994/.022	–	.936/.283	–	–	.734/.560
Feature denoising L3 (R18)	OOD+	Absolute improvement	.505/.930	.556/.892	.531/.911	.988/.053	.616/.836	.928/.353	.613/.989	.786/.558	.658/.734
Feature denoising L4 (R18)	ID+	Absolute improvement	.774/.825	–	–	.621/.981	–	.806/.760	–	–	.744/.848
Feature denoising L4 (R18)	OOD+	Absolute improvement	.769/.631	.780/.568	.775/.599	.716/.646	.758/.631	.784/.579	.743/.696	.750/.638	.762/.619
Mahalanobis [3]	OOD+	MDS	.559/.880	.615/.790	.587/.835	.675/.717	.631/.796	.707/.672	.763/.705	.694/.723	.640/.779
k-NN [5]	OOD+	KNN	.770/.728	.833/.496	.802/.612	.824/.486	.794/.607	.842/.517	.837/.536	.824/.536	.813/.574
PixMix [6]	OOD+	MSP	.766/.622	.832/.515	.799/.568	.696/.703	.814/.551	.934/.308	.918/.375	.841/.484	.820/.526
ReAct [4]	OOD+	ReAct	.787/.613	.829/.515	.808/.564	.784/.560	.800/.553	.830/.504	.802/.550	.804/.542	.806/.553
ASH [14]	OOD+	ASH	.765/.681	.799/.634	.782/.657	.772/.666	.788/.630	.856/.460	.807/.613	.806/.592	.794/.625
ViM [15]	OOD+	ViM	.722/.706	.778/.547	.750/.626	.819/.483	.758/.616	.831/.462	.859/.469	.817/.507	.783/.567

Table 15: CIFAR-100 as the ID dataset. Cells report AUROC/FPR@95 under the stated positive-class convention.

Variant	Conv.	Selected OOD score	Near-OOD			Far-OOD				Balanced
			SSB-hard	NINCO	Average	iNaturalist	Textures	OpenImage-O	Average	Macro
Teacher MSP	ID+	MSP	.804/.726	.862/.648	.833/.687	.931/.398	.883/.496	.892/.564	.902/.486	.867/.586
Teacher MSP	OOD+	MSP	.804/.660	.862/.445	.833/.552	.931/.257	.883/.453	.892/.353	.902/.355	.867/.454
Teacher energy	ID+	Energy	.798/.730	.853/.661	.826/.695	.932/.414	.906/.413	.896/.558	.911/.462	.868/.579
Teacher energy	OOD+	Energy	.798/.695	.853/.494	.826/.595	.932/.237	.906/.429	.896/.355	.911/.340	.868/.467
Affine pixels	ID+	Student energy	.791/.736	.844/.685	.817/.711	.921/.473	.907/.410	.889/.578	.906/.487	.862/.599
Affine pixels	OOD+	Student energy	.791/.712	.844/.512	.817/.612	.921/.270	.907/.414	.889/.382	.906/.355	.862/.484
Channel clipping	ID+	Student energy	.770/.760	.830/.695	.800/.728	.919/.450	.913/.389	.877/.605	.903/.481	.852/.604
Channel clipping	OOD+	Student energy	.770/.735	.830/.568	.800/.651	.919/.291	.913/.372	.877/.401	.903/.355	.852/.503
Spatial clipping	ID+	Energy gap	.797/.734	.852/.664	.825/.699	.931/.423	.905/.420	.895/.563	.910/.469	.868/.584
Spatial clipping	OOD+	Energy gap	.797/.695	.852/.495	.825/.595	.931/.238	.905/.429	.895/.355	.910/.341	.868/.468
Feature denoising, layer 2	ID+	Absolute improvement	.628/.896	.617/.843	.623/.869	.446/.984	.608/.782	.489/.915	.515/.894	.569/.882
Feature denoising, layer 2	OOD+	Absolute improvement	.628/.886	.617/.923	.623/.904	.446/.938	.608/.997	.489/.973	.515/.970	.569/.937
Feature denoising, layer 3	ID+	Absolute improvement	.642/.894	.695/.797	.668/.845	.745/.733	.707/.673	.621/.849	.691/.752	.680/.799
Feature denoising, layer 3	OOD+	Absolute improvement	.642/.848	.695/.826	.668/.837	.745/.766	.707/.978	.621/.909	.691/.885	.680/.861
Feature denoising, layer 4	ID+	Absolute improvement	.722/.797	.761/.755	.742/.776	.869/.556	.908/.367	.847/.607	.875/.510	.808/.643
Feature denoising, layer 4	OOD+	Absolute improvement	.723/.782	.762/.713	.742/.747	.871/.474	.908/.444	.847/.540	.875/.486	.809/.617
Mahalanobis [3]	OOD+	MDS	.584/.837	.655/.746	.619/.791	.750/.585	.792/.582	.699/.683	.747/.617	.683/.704
k-NN [5]	OOD+	KNN	.770/.737	.861/.466	.816/.602	.940/.245	.953/.244	.902/.329	.932/.273	.874/.437
PixMix [6]	OOD+	MSP	.788/.678	.855/.463	.822/.570	.925/.273	.898/.444	.884/.371	.902/.363	.862/.467
ReAct [4]	OOD+	ReAct	.790/.715	.848/.535	.819/.625	.937/.230	.929/.297	.904/.329	.923/.285	.871/.455
ASH [14]	OOD+	ASH	.795/.721	.852/.576	.824/.649	.951/.225	.948/.257	.918/.337	.939/.273	.881/.461
ViM [15]	OOD+	ViM	.740/.713	.833/.471	.787/.592	.910/.273	.946/.204	.882/.339	.913/.272	.850/.432

Table 16: Selected ResNet-18 variants and published OpenOOD baselines on ImageNet-200. Cells report AUROC/FPR@95 under the stated positive-class convention.

7 Conclusion

During this internship, we investigated several ways of using model distillation for OOD detection: classical output-level distillation, distillation under pixel- and embedding-level perturbations, ensembles of students trained on fixed feature subspaces, and feature denoising. The last direction departed from classical distillation by asking a student to recover a clean teacher representation from a corrupted one. Together, these experiments examined whether the information lost, distorted, or disagreed upon by a restricted student could provide a reliable indication that an input does not belong to the training distribution.

The experiments show that such signals do exist. Teacher-student mismatch, student confidence, ensemble disagreement, and feature reconstruction all separated ID and OOD samples to some extent. Feature denoising was particularly effective on several Far-OOD datasets, where the distribution shift is large. Output-level distillation was generally more competitive on Near-OOD datasets, where the student could still exploit class-related information. These results suggest that the two families capture different aspects of distribution shift: feature predictability is useful for large structural changes, whereas logits and probabilities remain more sensitive to changes close to the teacher’s decision boundary. This qualitative Near/Far split is consistent with the findings of Activation Subspaces [16]. That work reports that classifier-insignificant activation components are especially useful for Far-OOD detection, while decisive, class-related components are more useful for Near-OOD detection.

However, no student-based method consistently surpassed the teacher MSP baseline across CIFAR-10, CIFAR-100, and ImageNet-200. The main difficulty was Near-OOD detection, especially when ID and OOD samples shared similar visual and semantic structure. The tested distillation objectives did not preserve a sufficiently strong signal for these subtle differences, and their results also remained below several established OOD methods.

In the final discussion with my supervisor, we considered whether the results were promising enough to continue along the same path. We concluded that the current formulation was not working sufficiently well to justify only scaling up the same experiments and that this is a normal point in an exploratory research project: once the main hypothesis has been tested and its limitations become clear, the next step is usually to revise the hypothesis, modify the learning objective, or pivot toward the part of the approach that showed the strongest signal. Here, that could involve a more targeted study of Near-OOD failure cases, a different objective for preserving subtle representation-level differences, or a reformulation of feature denoising.

An additional and unexpected outcome of the internship concerned evaluation methodology. The positive-class convention used for FPR@95—ID positive or OOD positive—is still not standardized, despite the effort represented by the OpenOOD benchmark. Even papers published at the same recent conference can use different conventions. As demonstrated empirically in the OOD detection field, changing the convention can produce a substantially lower FPR@95 from exactly the same predictions, while AUROC remains unchanged. Without an explicit definition of the positive class and threshold population, this difference can be mistaken for a genuine improvement over another method.

References

- [1] G. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” 2015. <https://arxiv.org/abs/1503.02531>
- [2] D. Hendrycks and K. Gimpel, “A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks,” ICLR, 2017. <https://openreview.net/forum?id=Hkg4TI9x1>
- [3] K. Lee, K. Lee, H. Lee, and J. Shin, “A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks,” NeurIPS, 2018. <https://arxiv.org/abs/1807.03888>
- [4] Y. Sun, C. Guo, and Y. Li, “ReAct: Out-of-distribution Detection With Rectified Activations,” NeurIPS, 2021. <https://arxiv.org/abs/2111.12797>
- [5] Y. Sun, Y. Ming, X. Zhu, and Y. Li, “Out-of-Distribution Detection with Deep Nearest Neighbors,” ICML, 2022. <https://arxiv.org/abs/2204.06507>
- [6] D. Hendrycks, A. Zou, M. Mazeika, L. Tang, B. Li, D. Song, and J. Steinhardt, “PixMix: Dreamlike Pictures Comprehensively Improve Safety Measures,” CVPR, 2022. <https://arxiv.org/abs/2112.05135>
- [7] W. Liu, X. Wang, J. D. Owens, and Y. Li, “Energy-based Out-of-distribution Detection,” NeurIPS, 2020. <https://proceedings.neurips.cc/paper/2020/hash/f5496252609c43eb8a3d147ab9b9c006-Abstract.html>
- [8] Y. LeCun, “A Path Towards Autonomous Machine Intelligence,” 2022. <https://openreview.net/pdf?id=BZ5a1r-kVsf>
- [9] M. Assran et al., “Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture,” CVPR, 2023. <https://arxiv.org/abs/2301.08243>
- [10] M. Lafon, E. Ramzi, C. Rambour, and N. Thome, “Hybrid Energy Based Model in the Feature Space for Out-of-Distribution Detection,” ICML, 2023. <https://arxiv.org/abs/2305.16966>
- [11] N. Houlsby, F. Huszár, Z. Ghahramani, and M. Lengyel, “Bayesian Active Learning for Classification and Preference Learning,” 2011. <https://arxiv.org/abs/1112.5745>
- [12] J. Yang et al., “OpenOOD: Benchmarking Generalized Out-of-Distribution Detection,” NeurIPS, 2022. https://openreview.net/forum?id=gT6j4_tskUt
- [13] J. Zhang et al., “OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection,” 2023. <https://arxiv.org/abs/2306.09301>
- [14] A. Djurisić, N. Bozanić, A. Ashok, and R. Liu, “Extremely Simple Activation Shaping for Out-of-Distribution Detection,” ICLR, 2023. <https://arxiv.org/abs/2209.09858>
- [15] H. Wang, Z. Li, L. Feng, and W. Zhang, “ViM: Out-of-Distribution with Virtual-Logit Matching,” CVPR, 2022, pp. 4921–4930. https://openaccess.thecvf.com/content/CVPR2022/html/Wang_ViM_Out-of-Distribution_With_Virtual-Logit_Matching_CVPR_2022_paper.html
- [16] B. Zöngür, R. Hesse, and S. Roth, “Activation Subspaces for Out-of-Distribution Detection,” 2025. <https://arxiv.org/abs/2508.21695>
- [17] Z. Xu et al., “Mind the Way You Select Negative Texts,” CVPR, 2026. <https://github.com/ZhikangXu0112/InterNeg/blob/main/openood/evaluators/metrics.py>
- [18] Y. Ding et al., “Enhancing Out-of-Distribution Detection with Extended Logit Normalization,” CVPR, 2026. https://openaccess.thecvf.com/content/CVPR2026/papers/Ding_Enhancing_Out-of-Distribution_Detection_with_Extended_Logit_Normalization_CVPR_2026_paper.pdf
- [19] T. Denipitiyage et al., “RankOOD: Class Ranking-based Out-of-Distribution Detection,” CVPR, 2026. https://openaccess.thecvf.com/content/CVPR2026/papers/Denipitiyage_RankOOD_-_Class_Ranking-based_Out-of-Distribution_Detection_CVPR_2026_paper.pdf